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Batch: A-2
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Subject: Programming with Java

ASSIGNMENT No : 23

TITLE : Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

Problem Statement

Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

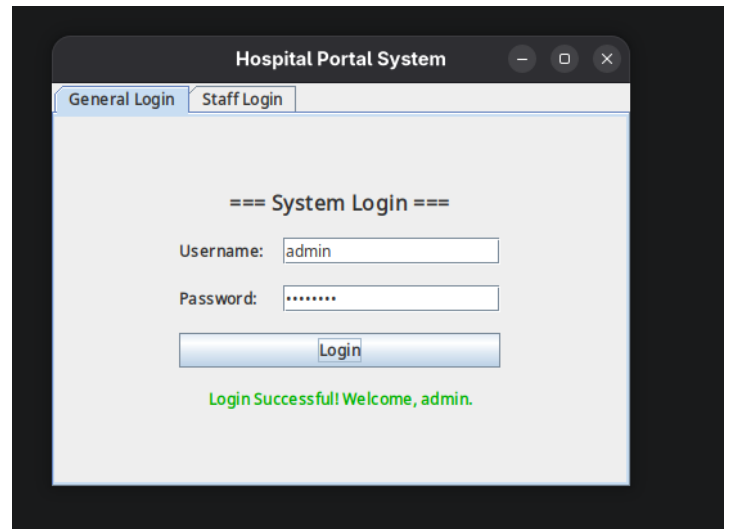
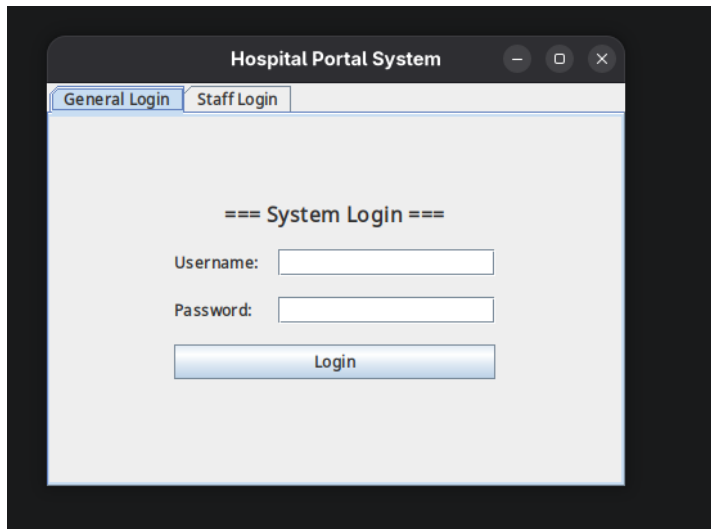
Learning Objectives

To navigate and retrieve records using ResultSet methods.

Theory

A `ResultSet` object contains data retrieved from a database after executing a `SELECT` query. It acts like a cursor that points to one row of the result at a time. Initially, the cursor is positioned before the first row. Various methods such as `next()`, `previous()`, `first()`, `last()`, `absolute()`, and `relative()` allow movement through the records. Data from each column can be retrieved using methods like `getString()`, `getInt()`, and `getDouble()`. Scrollable and updatable `ResultSets` provide greater flexibility for navigating and modifying retrieved data. `ResultSet` plays a crucial role in displaying and processing database information in Java applications.

Output Screenshots



CODE 1

```
1 package Exp23;
2
3 import javax.swing.*;
4 import java.awt.*;
5 import java.awt.event.ActionEvent;
6 import java.awt.event.ActionListener;
7 import java.sql.*;
8
9 public class Main extends JFrame {
10     private static final String DB_URL = "jdbc:mysql://localhost:3306/
hospital_db";
11     private static final String DB_USER = "root";
12     private static final String DB_PASSWORD = "devansh@2007";
13
14     public Main() {
15         setTitle("Hospital Portal System");
16         setSize(450, 350);
17         setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
18         setLocationRelativeTo(null);
19
20         JTabbedPane tabbedPane = new JTabbedPane();
21         tabbedPane.addTab("General Login", createGeneralLoginPanel());
22         tabbedPane.addTab("Staff Login", createStaffLoginPanel());
23
24         add(tabbedPane);
```

```

25     }
26
27     private JPanel createGeneralLoginPanel() {
28         JPanel panel = new JPanel(new GridBagLayout());
29         GridBagConstraints gbc = new GridBagConstraints();
30         gbc.insets = new Insets(8, 8, 8, 8);
31         gbc.fill = GridBagConstraints.HORIZONTAL;
32
33         JLabel lblTitle = new JLabel("=== System Login ===", SwingConstants.CEN
TER);
34         lblTitle.setFont(new Font("Arial", Font.BOLD, 16));
35
36         JLabel lblUser = new JLabel("Username:");
37         JTextField txtUser = new JTextField(15);
38         JLabel lblPass = new JLabel("Password:");
39         JPasswordField txtPass = new JPasswordField(15);
40         JButton btnLogin = new JButton("Login");
41         JLabel lblStatus = new JLabel("", SwingConstants.CENTER);
42
43         gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
44         panel.add(lblTitle, gbc);
45
46         gbc.gridwidth = 1;
47         gbc.gridx = 0; gbc.gridy = 1;
48         panel.add(lblUser, gbc);
49         gbc.gridx = 1; gbc.gridy = 1;
50         panel.add(txtUser, gbc);
51
52         gbc.gridx = 0; gbc.gridy = 2;
53         panel.add(lblPass, gbc);
54         gbc.gridx = 1; gbc.gridy = 2;
55         panel.add(txtPass, gbc);
56
57         gbc.gridx = 0; gbc.gridy = 3; gbc.gridwidth = 2;
58         panel.add(btnLogin, gbc);
59
60         gbc.gridx = 0; gbc.gridy = 4; gbc.gridwidth = 2;
61         panel.add(lblStatus, gbc);
62
63         btnLogin.addActionListener(new ActionListener() {
64             @Override
65             public void actionPerformed(ActionEvent e) {
66                 String username = txtUser.getText();
67                 String password = new String(txtPass.getPassword());

```

```

68
69         String query = "SELECT * FROM users WHERE username = ? AND pass
word = ?";
70
71         try (Connection conn = DriverManager.getConnection(DB_URL, DB_U
SER, DB_PASSWORD);
72
73             PreparedStatement pstmt = conn.prepareStatement(query)) {
74
75                 pstmt.setString(1, username);
76                 pstmt.setString(2, password);
77
78                 try (ResultSet rs = pstmt.executeQuery()) {
79                     if (rs.next()) {
80                         lblStatus.setForeground(Color.GREEN.darker());
81                         lblStatus.setText("Login Successful! Welcome, " + r
s.getString("username") + ".");
82                     } else {
83                         lblStatus.setForeground(Color.RED);
84                         lblStatus.setText("Invalid Username or Password.");
85                     }
86                 } catch (Exception ex) {
87                     lblStatus.setForeground(Color.RED);
88                     lblStatus.setText("Database Error: " + ex.getMessage());
89                 }
90             });
91
92         return panel;
93     }
94
95     private JPanel createStaffLoginPanel() {
96         JPanel panel = new JPanel(new GridBagLayout());
97         GridBagConstraints gbc = new GridBagConstraints();
98         gbc.insets = new Insets(8, 8, 8, 8);
99         gbc.fill = GridBagConstraints.HORIZONTAL;
100
101         JLabel lblTitle = new JLabel("=== Hospital Staff Portal ===", SwingCons
tants.CENTER);

```

```

102     lblTitle.setFont(new Font("Arial", Font.BOLD, 16));
103
104     JLabel lblStaffId = new JLabel("Staff ID:");
105     JTextField txtStaffId = new JTextField(15);
106     JLabel lblPass = new JLabel("Password:");
107     JPasswordField txtPass = new JPasswordField(15);
108     JButton btnLogin = new JButton("Login");
109     JLabel lblStatus = new JLabel("", SwingConstants.CENTER);
110
111     gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
112     panel.add(lblTitle, gbc);
113
114     gbc.gridwidth = 1;
115     gbc.gridx = 0; gbc.gridy = 1;
116     panel.add(lblStaffId, gbc);
117     gbc.gridx = 1; gbc.gridy = 1;
118     panel.add(txtStaffId, gbc);
119
120     gbc.gridx = 0; gbc.gridy = 2;
121     panel.add(lblPass, gbc);
122     gbc.gridx = 1; gbc.gridy = 2;
123     panel.add(txtPass, gbc);
124
125     gbc.gridx = 0; gbc.gridy = 3; gbc.gridwidth = 2;
126     panel.add(btnLogin, gbc);
127
128     gbc.gridx = 0; gbc.gridy = 4; gbc.gridwidth = 2;
129     panel.add(lblStatus, gbc);
130
131     btnLogin.addActionListener(new ActionListener() {
132         @Override
133         public void actionPerformed(ActionEvent e) {
134             String staffId = txtStaffId.getText();
135             String password = new String(txtPass.getPassword());
136
137             String query = "SELECT role FROM hospital_staff WHERE staff_id
= ? AND password = ?";
138
139             try (Connection conn = DriverManager.getConnection(DB_URL, DB_U
SER, DB_PASSWORD);
140
141                 PreparedStatement pstmt = conn.prepareStatement(query)) {

```

```

142         pstmt.setString(2, password);
143
144         try (ResultSet rs = pstmt.executeQuery()) {
145             if (rs.next()) {
146                 String role = rs.getString("role");
147                 lblStatus.setForeground(Color.GREEN.darker());
148
149                 if ("Doctor".equalsIgnoreCase(role)) {
150
151                     lblStatus.setText("Access Granted: Welcome, Dr.
152
153                         " + staffId);
154
155                     } else if ("Nurse".equalsIgnoreCase(role)) {
156
157                         lblStatus.setText("Access Granted: Welcome, Nur
158
159                             se " + staffId);
160
161                     } else {
162
163                         lblStatus.setText("Access Granted: Welcome, Sta
164
165                             ff member.");
166                     }
167                 } else {
168                     lblStatus.setForeground(Color.RED);
169
170                     lblStatus.setText("Authentication Failed: Invalid S
171
172                         taff ID or Password.");
173                 }
174             } catch (Exception ex) {
175                 lblStatus.setForeground(Color.RED);
176
177                 lblStatus.setText("Database Error: " + ex.getMessage());
178             }
179         });
180
181         return panel;
182     }
183
184     public static void main(String[] args) {
185         SwingUtilities.invokeLater(() -> {
186             new Main().setVisible(true);
187         });
188     }
189 }

```

EXERCISE 1

```
1 package Exp23.Exercise;
2
3 import java.sql.*;
4
5 public class Exp1 {
6     public static void main(String[] args) {
7         try {
8             Connection con = DriverManager.getConnection(
9                 "jdbc:mysql://localhost:3306/hospital_db",
10                 "root",
11                 "devansh@2007"
12             );
13
14             Statement stmt = con.createStatement();
15             ResultSet rs = stmt.executeQuery("SELECT * FROM
hospital_staff");
16
17             while (rs.next()) {
18                 System.out.println(
19                     "Staff ID: " + rs.getString("staff_id") +
20                     ", Role: " + rs.getString("role")
21                 );
22             }
23
24             con.close();
25         } catch (Exception e) {
26             System.out.println("Error: " + e.getMessage());
27         }
28     }
29 }
```

EXERCISE 2

```
1 package Exp23.Exercise;
2
3 import java.sql.*;
4
5 public class Exp2 {
6     public static void main(String[] args) {
7         try {
8             Connection con = DriverManager.getConnection(
9                 "jdbc:mysql://localhost:3306/hospital_db",
10                 "root",
```

```

11         "devansh@2007"
12     );
13
14     Statement stmt = con.createStatement();
15     ResultSet rs = stmt.executeQuery("SELECT * FROM employee");
16
17     while (rs.next()) {
18         System.out.println(
19             "ID: " + rs.getInt("emp_id") +
20             ", Name: " + rs.getString("name") +
21             ", Department: " + rs.getString("department") +
22             ", Salary: " + rs.getDouble("salary")
23         );
24     }
25
26     con.close();
27 } catch (Exception e) {
28     System.out.println("Error: " + e.getMessage());
29 }
30 }
31 }

```

SQL CODE:

```

1  -- Database: hospital_db
2  CREATE DATABASE hospital_db;
3  USE hospital_db;
4
5  -- Table for Exercise 1
6  CREATE TABLE users (
7      id INT PRIMARY KEY AUTO_INCREMENT,
8      username VARCHAR(50) NOT NULL,
9      password VARCHAR(50) NOT NULL
10 );
11 INSERT INTO users (username, password) VALUES ('admin', 'admin123');
12
13 -- Table for Exercise 2
14 CREATE TABLE hospital_staff (
15     id INT PRIMARY KEY AUTO_INCREMENT,
16     staff_id VARCHAR(50) UNIQUE NOT NULL,
17     password VARCHAR(50) NOT NULL,
18     role VARCHAR(20) NOT NULL -- 'Doctor' or 'Nurse'
19 );
20 INSERT INTO hospital_staff (staff_id, password, role) VALUES
21 ('DOC101', 'docpass', 'Doctor'),

```

```

22 ('NUR201', 'nurpass', 'Nurse');
23
24 CREATE TABLE employee (
25     emp_id INT,
26     name VARCHAR(50),
27     department VARCHAR(50),
28     salary DOUBLE
29 );
30
31 INSERT INTO employee VALUES
32 (101, 'Rahul', 'IT', 50000),
33 (102, 'Amit', 'HR', 45000),
34 (103, 'Neha', 'Finance', 55000);
35

```

Exercise

Q1. Create a program to display database records one by one using ResultSet.

```

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp23/Exercise/Exp1.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp23.Exercise.Exp1
Staff ID: DOC101, Role: Doctor
Staff ID: NUR201, Role: Nurse
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$

```

Q2. Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department, and salary sequentially.

```

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp23/Exercise/Exp2.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp23.Exercise.Exp2
ID: 101, Name: Rahul, Department: IT, Salary: 50000.0
ID: 102, Name: Amit, Department: HR, Salary: 45000.0
ID: 103, Name: Neha, Department: Finance, Salary: 55000.0
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$

```

Conclusion

Thus, the experiment successfully demonstrated how to use `ResultSet` in JDBC to navigate and retrieve records from a database. Various `ResultSet` methods were used to move through records and retrieve column values, providing an understanding of how database information can be accessed and displayed in Java applications.